# Parallel Travelling Salesman Problem Solver

## Island-Model Genetic Algorithm with MPI

This report describes a parallel solver for the
Travelling Salesman Problem
(TSP). The implementation uses an island-model Genetic
Algorithm (GA) written in
C++ and parallelized with MPI. Each MPI process is an
independent island that
evolves a population of tours. At fixed synchronization
intervals, all islands
cooperate by finding the global best tour with
MPI_Allreduce and
broadcasting the owner tour with MPI_Bcast. Other
islands do not clone
the whole tour; instead, they splice random contiguous
segments of the global
best into their own population through order crossover.
This keeps the search
diverse while still spreading useful sub-routes.
The repository also contains Python tools for
visualization, validation, and
plot generation. The Python code does not implement the
TSP algorithm; it only
drives the C++ executable or reads output files
generated by the solver. The
experimental results cover correctness validation,
input-size selection,
granularity/load balance, and speedup with and without
communication time.

# Project Overview

## Problem Statement

Given a set of N cities with two-dimensional
coordinates, the Travelling
Salesman Problem asks for a closed tour that visits
every city exactly once and
returns to the starting city while minimizing total
Euclidean distance. For a
tour T = (t_0, t_1, , t_N-1), the objective value is:
L(T) = _i=0^N-1 d(t_i, t_(i+1) N)
where d(a,b) is the Euclidean distance between city a
and city b.
TSP is NP-hard, so the project uses a heuristic search
method instead of exact
enumeration for large instances.

## Repository Structure

The project is organized as follows:
sequential baseline, and unit tests.

orchestration tools.
cluster launch scripts.

### Main Source Files

The full solver is in C++:
evaluates tours, creates random tours, performs tournament selection, order
crossover (OX), mutation, and one generation of evolution.
unit tests for genetic operators and local search.

## Sequential Genetic Algorithm

### Representation

Each individual is a permutation of city indices. For example, for N=5, a
tour may be:
The tour is closed implicitly, so the last city connects back to the first city.
This representation guarantees that a valid individual can be evaluated directly
as a TSP route.

### Fitness Evaluation

The solver precomputes a flat N N Euclidean distance matrix. The
fitness value is the total tour length. Smaller values are better.

### Selection, Crossover, and Mutation

The GA uses:
and fills the remaining positions using the order of the second parent.

### Local Search

The optional memetic extension applies local search to the best individual every
segment.

## Parallelization Design

### Parallelism Level

The implementation uses coarse-grained task-level parallelism. Each MPI process
owns a complete GA population and performs an independent stochastic search.
The city data and distance matrix are replicated on every process.
The design also has an exploratory-search characteristic: different islands
explore different regions of the search space due to different random seeds.
Therefore, the decomposition is best described as a hybrid of task decomposition

and exploratory decomposition.

## Decomposition Technique

The selected decomposition is exploratory decomposition with replicated
input data. Each island receives the same TSP instance but evolves a different
population. This is suitable for a heuristic
optimization problem because the
main work is search, not deterministic matrix
processing.
The alternatives were less suitable:
the global order of all cities.
search, not for the selected GA.
not the focus of this project.

## Mapping Technique

The mapping is a one-dimensional process-to-island assignment:
Each rank stores:
For the cluster experiment, the hostfile uses a
sequential round-robin list of
hosts. This spreads ranks across four machines and supports up to 48 ranks, with
12 ranks per machine as the common physical-core limit.

## Communication Strategy

Communication is periodic and collective. Every -sync generations:
length and the owner rank.
global best with local individuals.
This is a blocking collective communication strategy.
The logical topology is
global collective communication rather than a ring or master-slave topology.
There is no permanent master rank during evolution; all ranks participate
symmetrically. Rank 0 is used only for final reporting
and file output.

## Migration Policy

A naive island GA can broadcast the full best
individual and clone it into every
island. That is fast but dangerous: diversity collapses and all islands may
converge to the same local optimum.
This implementation uses partial global-best migration:
random local mate.
Only sub-routes are shared. Random cut points and local mates keep the resulting
population different on each island.

## Load Balancing Considerations

In the default mode, every rank uses the same
population size, so compute work is
approximately balanced. This is appropriate for
machines with similar effective
capacity or for a controlled experiment where ranks per
machine are limited by
the slowest machine.
The solver also contains an optional -auto-balance
mode. It performs a
small local benchmark on each process, gathers the
measured times, and assigns a
larger population to faster ranks. The allocation is
clamped between 0.5x and 2x
of the original population to avoid unstable decisions
from noisy measurements.

### Early Stopping

If -patience is enabled, the solver stops when the
global best does not
improve for a fixed number of generations. Because the
same global best value is
known to all ranks after each synchronization, every
rank can make the same stop
decision without an extra control message.

## Parallel Algorithm Pseudocode

Pseudocode / command listing omitted in preview PDF.

## Implementation and Instrumentation

### Command-Line Interface

The main solver accepts the following key arguments:
best per synchronization.

### Timing Metrics

The solver separates communication time from total
elapsed time:
These metrics are written to the -stats CSV file and
used by

### Cluster Environment

The cluster setup uses four machines and up to 48 MPI
ranks. OpenMPI 5.0.9 is
source-built and pinned at /opt/openmpi-5.0.9 on all
nodes to avoid
runtime incompatibilities. The launch script uses:
heterogeneous machines.

## Correctness Validation

### Permutation Validity

The solver represents a route as a permutation of city
IDs. The unit tests check
that crossover and mutation preserve valid

permutations. The validation script

## Small-Instance Exact Check

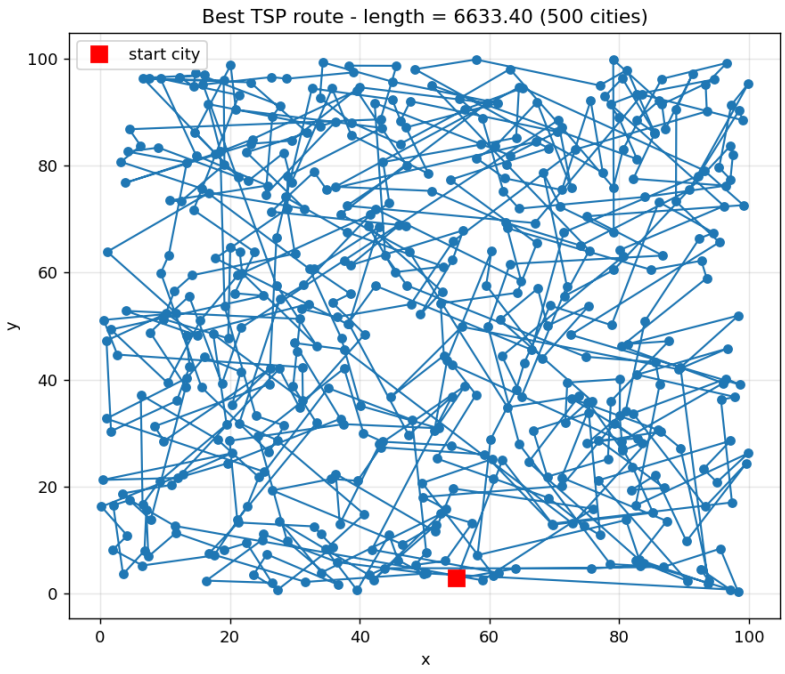
For N 10, the validation script can brute-force all tours with city 0 fixed as the first city. This gives the true optimum and allows the GA result to be checked against the exact solution. The repository includes validation logs for small and larger instances in results/.

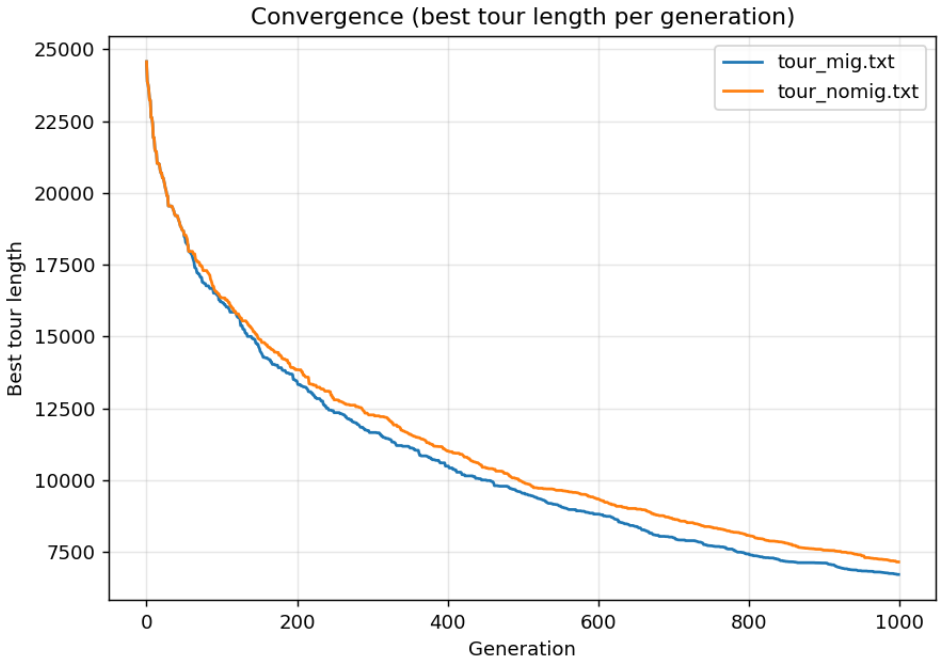
## Parallel Communication Check

The per-rank history files show that migration changes the behavior of non-owner islands after synchronization points. In one cluster run, rank 2 held the best tour before synchronization at generation 20. After the synchronization, weaker islands moved toward that better solution, but did not become identical. This matches the intended partial-migration behavior: useful route segments spread, while population diversity is preserved.

## Visualization Outputs

The repository includes route and convergence visualizations generated by output route is a valid closed TSP tour and that the best length decreases over the generations.



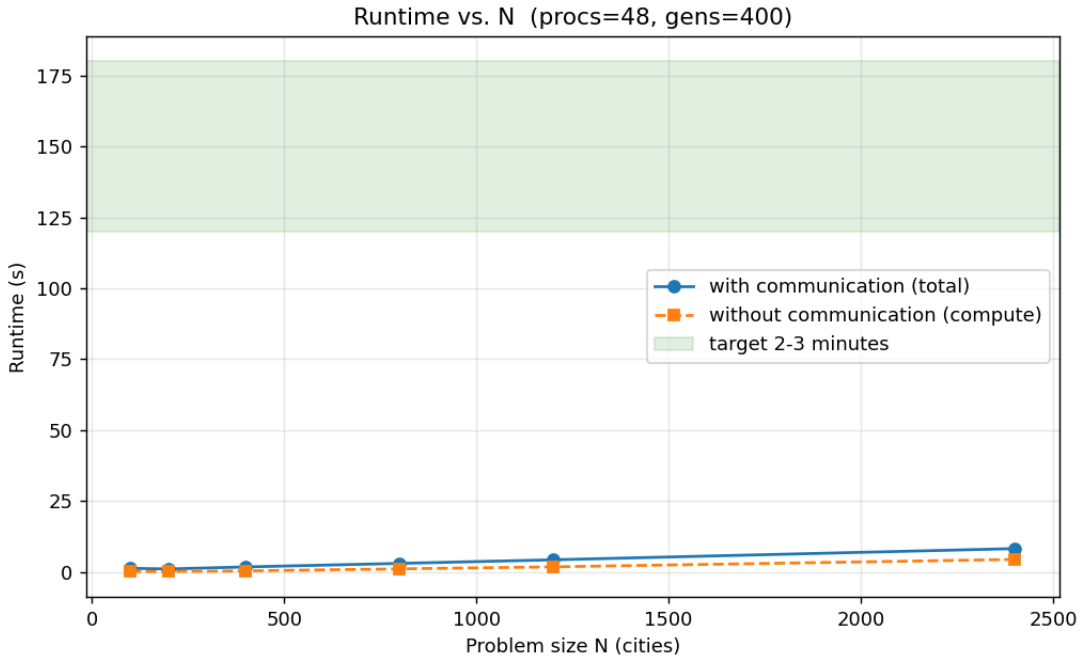



# Experimental Results

## Experiment 1: Input Size Selection

The requirement asks for an input-size experiment using the number of processes
equal to the number of physical CPU cores. The cluster configuration uses up to
48 ranks. The runtime is measured in two cases: total time including
communication, and estimated compute time excluding communication.
The first sweep used a fixed 400-generation run to understand the growth trend.
It showed that larger instances increase compute work and communication volume,
but the run was still too short for the required 2-3 minute target.

N | Total time (s) | Communication (s) | Compute only (s)
100 | 1.2427 | 1.1512 | 0.0915
200 | 1.0223 | 0.8770 | 0.1453
400 | 1.7410 | 1.4027 | 0.3383
800 | 2.9824 | 1.9272 | 1.0552
1200 | 4.2491 | 2.4806 | 1.7685
2400 | 8.1918 | 3.8092 | 4.3826




To satisfy the 2-3 minute requirement, the final selection fixed

N | Total time (s) | Communication (s) | Compute only (s)

1200 | 54.54 | 25.90 | 28.65

1800 | 86.99 | 34.33 | 52.66

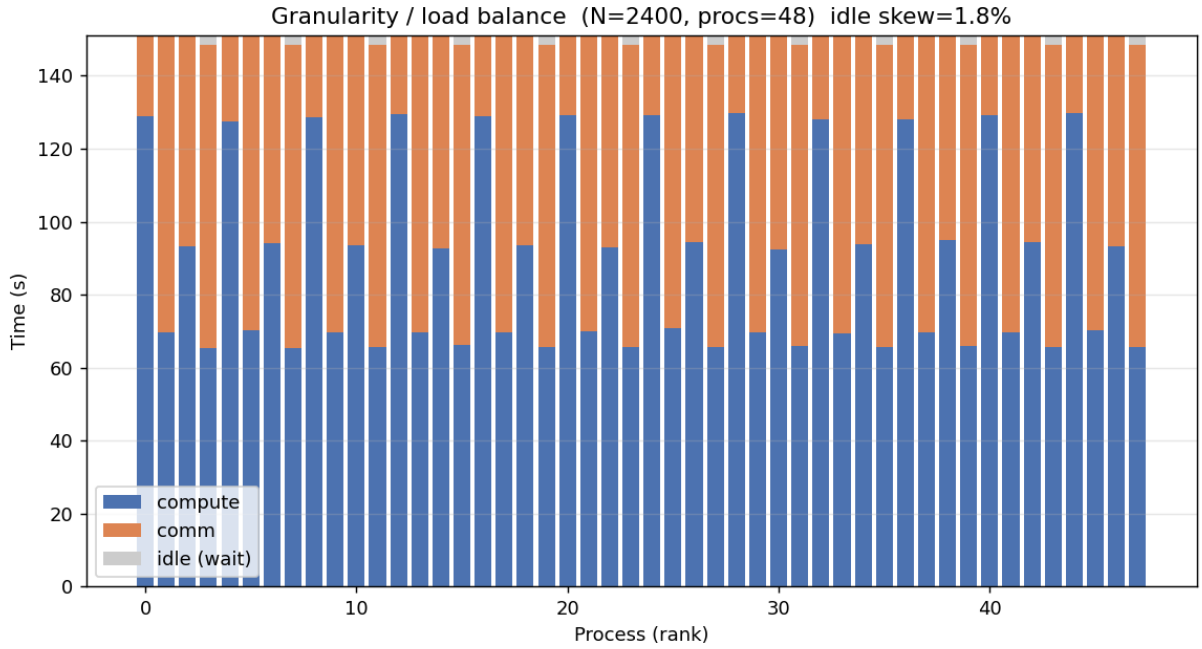
2400 | 154.27 | 64.05 | 90.22

The selected input size is therefore:

The official repeated run stored in results/nstar_run.log reports a makespan of 145.97 seconds, which lies inside the required 120-180 second interval. Other repeated measurements at the same configuration were 159.14s and 154.27s, also inside the target range.

## Experiment 2: Granularity and Load Balance

For the granularity experiment, the program is run with the selected input size N^*=2400, gens=10500, and 48 ranks. Each rank is one island, so the granularity is one GA population per process. The required stacked bar chart shows compute time and communication time in the same column for each process.

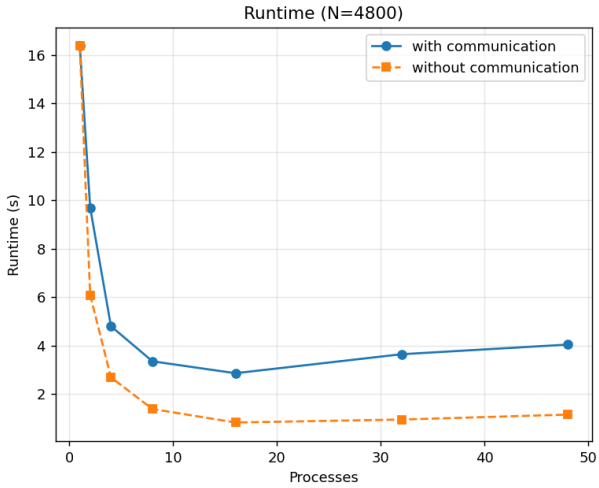


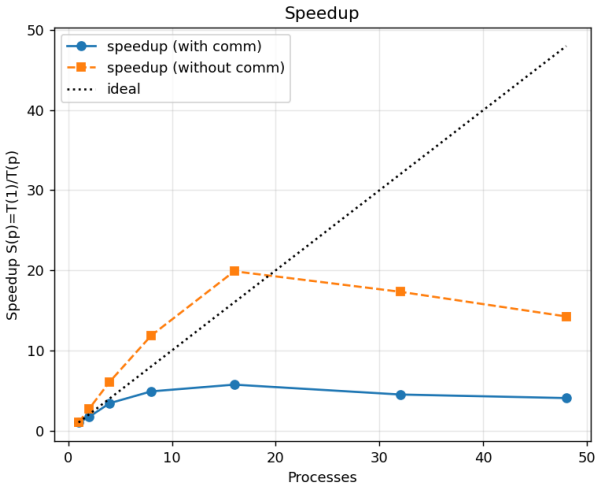

The measured idle skew is about 1.8\
threshold. Therefore, the system is considered load-balanced for this run. The
main difference between ranks is not idle waiting but the split between compute
and communication time, which is expected when ranks synchronize through
blocking MPI collectives.

## Experiment 3: Speedup

The speedup experiment uses a fixed total workload and varies the number of
processes. Following the requirement, this experiment uses 2N^* = 4800 cities
with gens=400, sync=20, and a fixed total population of 480
individuals split over the active ranks. The measured values are:

Procs | Total (s) | Comm (s) | Compute (s) | Speedup | No-comm speedup | Eff.
1 | 16.3636 | 0.0000 | 16.3636 | 1.0000 | 1.0000 | 1.0000
2 | 9.6910 | 3.6071 | 6.0840 | 1.6885 | 2.6896 | 0.8443
4 | 4.7995 | 2.1130 | 2.6865 | 3.4094 | 6.0910 | 0.8524
8 | 3.3492 | 1.9690 | 1.3802 | 4.8858 | 11.8559 | 0.6107
16 | 2.8577 | 2.0346 | 0.8231 | 5.7261 | 19.8806 | 0.3579
32 | 3.6413 | 2.6965 | 0.9448 | 4.4939 | 17.3200 | 0.1404
48 | 4.0388 | 2.8897 | 1.1491 | 4.0516 | 14.2398 | 0.0844




The speedup improves up to 16 processes and then decreases. This means the parallel computation scales initially, but the communication and synchronization overhead becomes dominant at high process counts. The no-communication speedup is much higher, showing that the GA computation itself is parallelizable. The limiting factor is the cost of repeated collective communication on the real cluster network.

# Discussion

## Why the Island Model Fits TSP

The island model is a natural fit for TSP because the problem is a global combinatorial search. Multiple populations can explore different areas of the solution space without requiring communication at every generation. This reduces the cost compared with fine-grained parallel GA designs.

## Communication Trade-Off

The -sync parameter controls the trade-off between cooperation and overhead. A small interval spreads good information quickly but increases communication cost and can reduce diversity. A large interval reduces overhead but makes the islands behave more like independent runs. The no-sharing mode

## Granularity

The chosen granularity is coarse: one full GA population per process. This reduces communication frequency and makes the program

easier to run on a
heterogeneous cluster. The drawback is that very small
TSP instances do not have
enough computation per process, so communication
dominates.

### Limitations

The current implementation is a heuristic solver. It
does not guarantee the
global optimum for large TSP instances. The exact check
is only feasible for
small N. The selected N^*=2400 configuration satisfies
the 2-3 minute
runtime target, but the result is still affected by
normal run-to-run variation
on a heterogeneous laptop cluster.

## Conclusion

This project implements a complete MPI-based parallel
TSP solver using an
island-model Genetic Algorithm. The parallelism is
coarse-grained and
task/exploratory in nature. Each MPI rank owns an
island and communicates only
periodically through blocking MPI collectives. The
mapping is one-dimensional:
rank to island. The communication topology is global
collective communication
using MPI_Allreduce and MPI_Bcast.
Correctness is checked through permutation tests,
independent route-length
validation, and brute-force validation for small
instances. The experimental
results show balanced work distribution and meaningful
speedup up to a moderate
number of processes. At high process counts,
communication overhead dominates,
which is visible in the difference between total
speedup and no-communication
speedup.

## Appendix: Reproducible Commands

Pseudocode / command listing omitted in preview PDF.